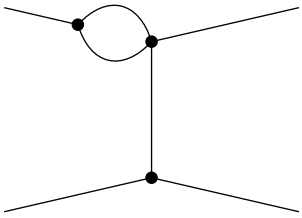
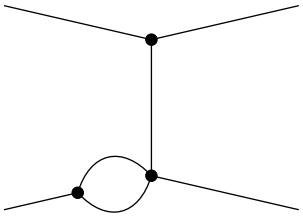
T-tree swordfish: direct shared vertex

Plain lines; no caps, circles, field labels or momenta.

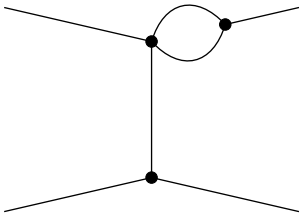
Slot 1



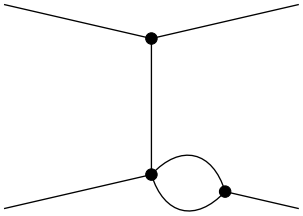
Slot 2



Slot 3



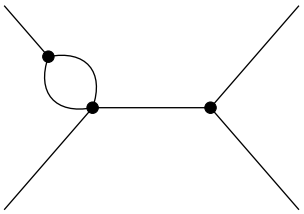
Slot 4



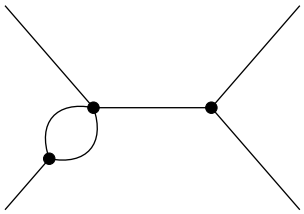
S-tree swordfish: direct shared vertex

Plain lines; no caps, circles, field labels or momenta.

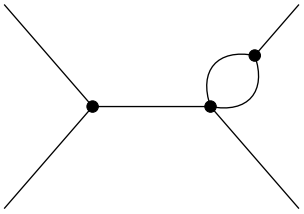
Slot 1



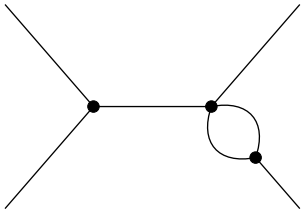
Slot 2



Slot 3



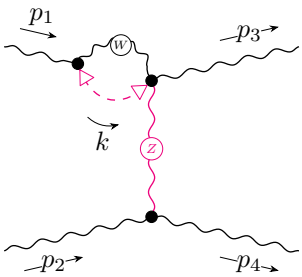
Slot 4



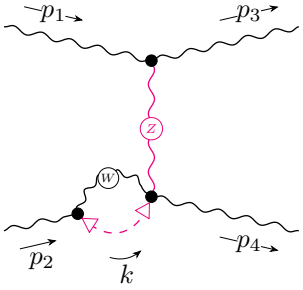
T-tree swordfish: momentum and mixed styles

One selected external guide; two independent loop edges into the actual tree vertex.

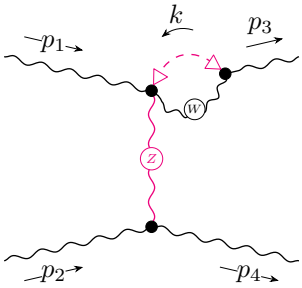
Slot 1



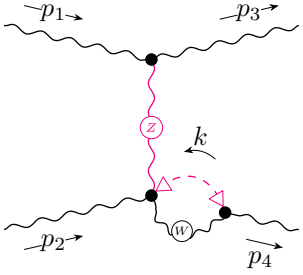
Slot 2



Slot 3



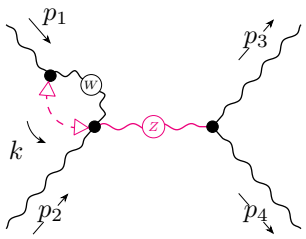
Slot 4



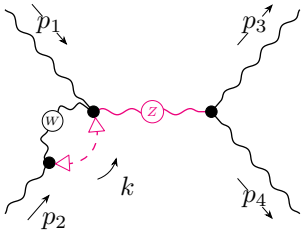
S-tree swordfish: momentum and mixed styles

One selected external guide; two independent loop edges into the actual tree vertex.

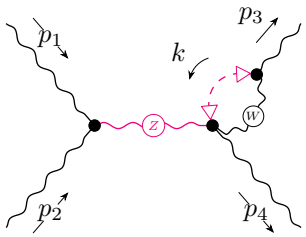
Slot 1



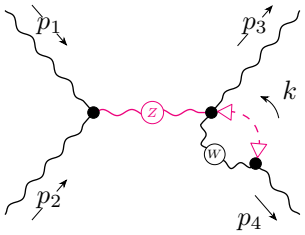
Slot 2



Slot 3



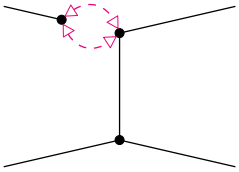
Slot 4



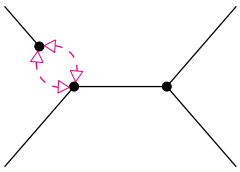
Proca endpoint contact: all eight placements

Only loop endpoint markers; plain host/external lines. No momentum, caps or circles.

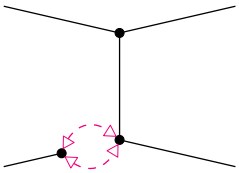
T slot 1



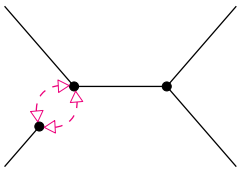
S slot 1



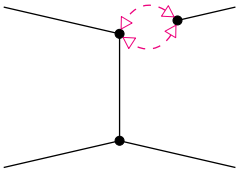
T slot 2



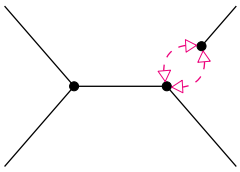
S slot 2



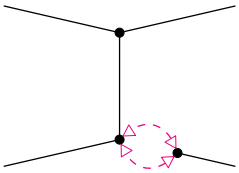
T slot 3



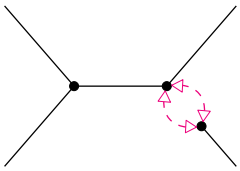
S slot 3



T slot 4



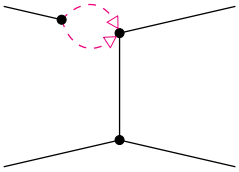
S slot 4



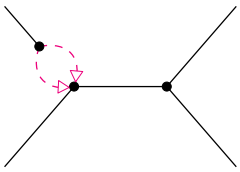
Scalar endpoint contact: all eight placements

Only loop endpoint markers; plain host/external lines. No momentum, caps or circles.

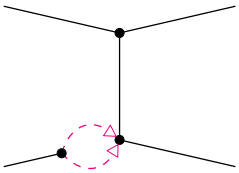
T slot 1



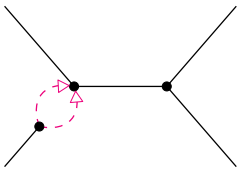
S slot 1



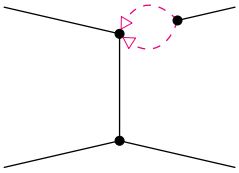
T slot 2



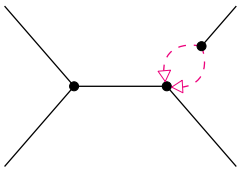
S slot 2



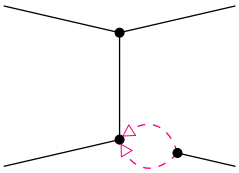
T slot 3



S slot 3



T slot 4



S slot 4

